

Ch 1. Sequence alignment algorithms

global optimal: $O(M*N)$ [quadratic]

simple but effective algorithms

define alignment & score of alignment

recursive algorithm, sequence combinatorics: prefixes, suffixes

1.1 Edit Graph

1.2 Scoring alignments, scoring scheme

δ	-	A	T	G	C
-	0	0	0	0	0
A	0	1	0	0	0
T	0	0	1	0	0
G	0	0	0	1	0
C	0	0	0	0	1

unit scoring scheme

pairwise sequence alignment

$X = x_1, x_2, \dots, x_n$

$Y = y_1, y_2, \dots, y_m$

ex) $X = A \ T \ T \ A \ C \ G$
 $Y = A \ T \ A \ T \ C \ G$ score = 4

$X = A \ T \ T \ A \ - \ C \ G$
 $Y = A \ - \ T \ A \ T \ C \ G$ score = 5

$x_i, y_j \in \Sigma = \{A, C, G, T\}$

$\delta(x_i, y_j)$
 letter w/ letter

$\delta(x_i, -)$ $\delta(-, y_j)$
 letter w/ dash

Edit Graph $G_{x,y}$: edge labelled directed graph

$G(V, E)$ V = vertex set E = edge set

Vertices: $(m+1) \times (n+1)$

start from 0
 $(0, m)$ $(0, n)$

(i, j) row i column j

edges: fundamental units of alignment

gap in Y , gap in X , alignment edge

gap in Y :
 $(i-1, j) \xrightarrow{(x_i, -)} (i, j)$
 gap in X :
 $(i, j-1) \xrightarrow{(-, y_j)} (i, j)$

alignment edge:

$(i-1, j-1) \xrightarrow{(x_i, y_j)} (i, j)$
 edged label

conclusion: there are exactly ≥ 1 edges going into vertex (i, j)

thm: there is a one-to-one correspondence b/w directed path from $(0, 0)$ to (M, N) and alignments b/w X and Y

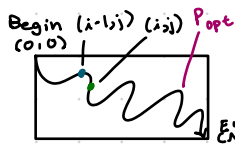
* any alignment just a direct path from start $\rightarrow$ end *

DAG: directed, acyclic graphs

→ solved w/ linear time $O(N)$

Dynamic Programming → useful for solving certain g's

→ Principle of optimality. Optimal paths are optimal between any 2 vertices



def. denote node $S(i,j)$ the score of the maximum directed path from $(0,0)$ to vertex (i,j)

because there are only 3 edges directed into (i,j) , it follows that any optimal (maximal) path P_{opt} to (i,j) must use one of these edges:

P_{opt} first arrives at...

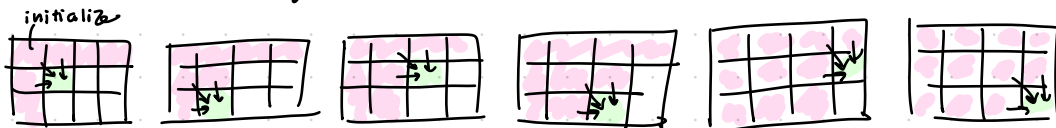
- ① $(i-1,j)$ follows gap edge in y to (i,j) , OR
- ② $(i,j-1)$ follows gap edge in x to (i,j) , OR
- ③ $(i-1,j-1)$ follows aligned edge to (i,j)

$$opt. path \quad S(i,j) = \text{MAX} \begin{cases} S(i-1,j-1) + \delta(x_i, y_j) \\ S(i-1, j) + \delta(x_j, -) \\ S(i, j-1) + \delta(-, y_j) \end{cases}$$

opt path b/f (i,j)
optimal path to (i,j)

We need to compute $S(i,j)$ [Optimal cost] such that @ any time, we have already computed the values of S needed by the recurrence (atleast 1 indices smaller than (i,j))

- there are several ways to do such a computation
- we will do it column by column.



 known S computed S → incoming edges

prove optimality via assuming the opposite

The global alignment algorithm

$$S[0,0] = 0$$

$$O(M*N) \approx O(N^2)$$

for $j=1$ to N :

$$S[0,j] = S[0,j-1] + \delta(-, y_j)$$

columns first $\downarrow$

for $i=1$ to M :

$$S[i,0] = S[i-1,0] + \delta(x_i, -)$$

shift row $\rightarrow$

for $j=1$ to N :

$$S[i,j] = \max \begin{bmatrix} S[i-1,j-1] + \delta(x_i, y_j) \\ S[i-1,j] + \delta(x_i, -) \\ S[i,j-1] + \delta(-, y_j) \end{bmatrix}$$

calculate max

print ("max score" is $S[M,N]$)

0,0			
0,1			
0,2			

columns first

$\rightarrow$ shift row

0,0	1,0		
0,1			
0,2			

calc max score

0,0	1,0	
0,1	2,1	
0,2		

Margaret Dayhoff

Window of non-conservation:

W W Y I R
W F Y V R
W Y Y V R
W Y Y T R
W F Y K R
W Y Y V R

(ungapped)
multiple
alignment

conserved

how far away from random?

B A B A
A A A C
A A C C
A A B A
A A C C
A A B C

There are 24 AAs.

14 A 14/24

4 B 4/24

6 C 6/24

$$4 * \binom{6}{2} = 4 * \frac{6*5}{2!} = 60 \text{ pairs aligned}$$

observed
frequency
of individual
AAs

aligned Pair

freq. observed

A - A

26/60

A - B

8/60

A - C

10/60

B - B

3/60

B - C

6/60

C - C

7/60

frequency
of
aligned
pairs

compared to observed

to expected frequencies

calculate random freqs
based on individual AA

probabilities

next page

A: $14/24$ We choose at random sequences with same frequency of AAs given
 B: $4/24$ and make an alignment. In a random alignment, the expected
 C: $6/24$ proportions of pairs is:

$$\begin{array}{l}
 A-A: \left(\frac{14}{24}\right)^2 \\
 A-B: \left(\frac{14}{24}\right)\left(\frac{4}{24}\right) \cdot 2 \\
 A-C: \left(\frac{14}{24}\right)\left(\frac{6}{24}\right) \cdot 2 \\
 B-B: \left(\frac{4}{24}\right)^2 \\
 B-C: \left(\frac{4}{24}\right)\left(\frac{6}{24}\right) \cdot 2 \\
 C-C: \left(\frac{6}{24}\right)^2
 \end{array}
 \left[\begin{array}{l} \text{EXPECTED} \\ \text{PROPORTIONS} \\ \text{OF PAIRS} \end{array} \right]$$

← B/C it can be A-C or C-A

everything is hypothesis testing:

estimated likelihood ratios: $2 \log_2 \left(\frac{\text{OBSERVED}}{\text{EXPECTED}} \right)$

aligned pair	obs	expected	$2 \log_2 \left(\frac{\text{obs}}{\text{expected}} \right)$	Scoring scheme
A-A	26/60	196/576	0.7	
A-B	8/60	112/576	-1.09	
A-C	10/60	168/576	-1.61	
B-B	3/60	16/576	1.70	
B-C	6/60	48/576	0.93	
C-C	7/60	36/576	1.83	

A B C
 A 1 -1 -2
 B -1 2 1
 C -2 1 2